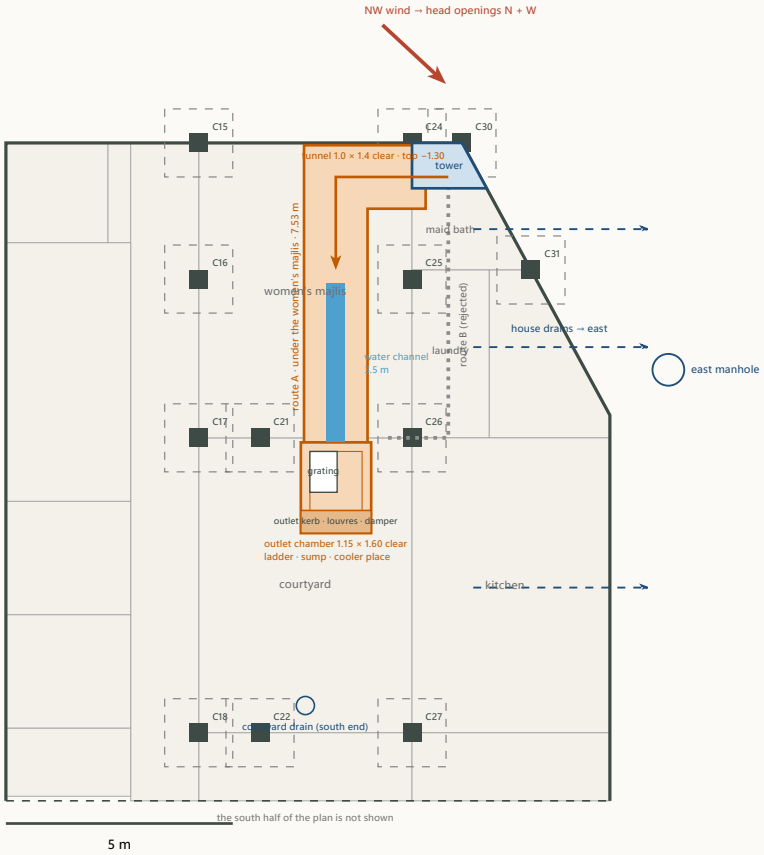
Air system study · malqaf, tunnel, water channel · Taif / Palm House V6.2

design-development study · every size, level and performance figure ASSUMED, to be checked by the MEP and structural engineers · CSV coordinates, north up

Nothing here is calculated: no airflow, cooling, structural or hydraulic figure. ALL CHECKS PASSED means the drawings are consistent with the project files, not that the system was sized.

Plan at foundation level · the tower, the tunnel (route A), the outlet chamber · assumed pads dashed



The idea

- The **malqaf** (wind tower) in the tip of the north-east cut corner catches the north-west wind above the roof, where it is strong and clean, and sends it **down the shaft** (≈ 0.88 m² clear) into an **underground tunnel** that runs through the cool ground to the courtyard. In the last metres the air passes over a **shallow water channel** (evaporative cooling and a dust trap) and enters the courtyard through a **low, closable outlet** in the north wall. Warm air leaves through the open sky. Every opening closes: dust storms, cold nights, hot afternoons.
- **Goal:** cools well, clean and healthy, easy to clean, and does not touch the house: a separate waterproof box, no standing water, no sprays, no route under bedrooms or wet rooms.

Route: two candidates, one chosen

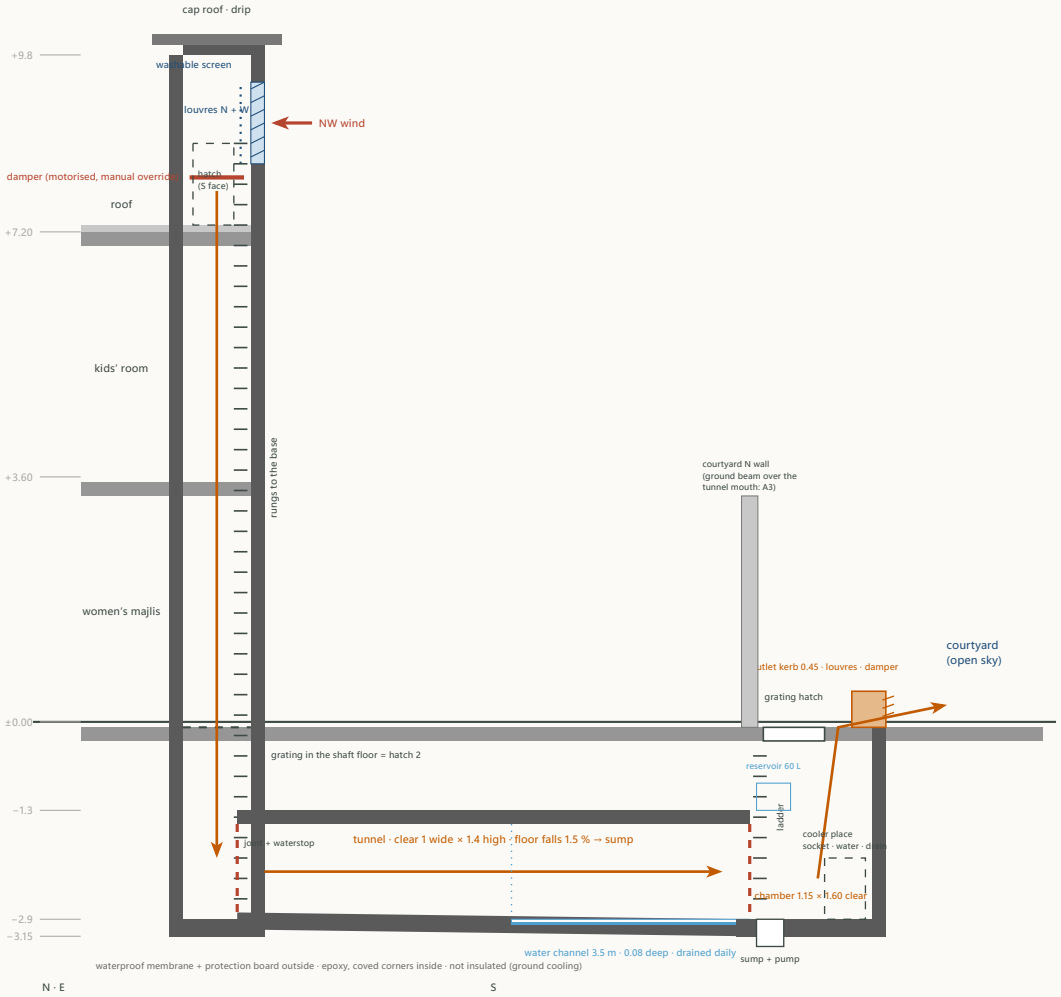
	A · under the women's majlis (chosen)	B · under the maid wing (rejected)
path	tower base → 1.68 m west under the majlis north strip → 5.85 m south on x 7.27 → outlet chamber at the courtyard's north wall; axis 7.53 m from the tower wall	tower base → south under the maid bath and the laundry (x ≈ 9.75) → west under the kitchen north wall to the courtyard corner; ≈ 8.5 m
above it	ground floor: the women's majlis only (a sitting room, no drains); first floor: the kids' room / family living. **No bedroom, no wet room.**	maid bath and laundry (wet rooms: drains, smells, leaks), then the kitchen north wall
footings crossed	the W3 / C24 line at the tower exit (unavoidable: the tower base box becomes their foundation, A1); the women's majlis south wall ground beam at the chamber (A3). Clear of the pads C25, C26, C21 (≥ 0.15 m, with the assumed 1.5 m pads)	the maid bath / laundry walls y 1.0, 2.8; the RC wall W5 (y 6.5) and the pad of C26 at the corner: three crossings
drains	none: the east block drains run east, away from the route	crosses the maid bath and laundry drains that run east
cleaning	one straight leg; the chamber gives the main access from the courtyard	two bends, longer, under wet rooms

Levels and sizes (assumed)

- **Head:** shaft walls from the roof finish +7.3 to +9.8 (2.5 m above the roof); louvred openings on the north and west faces, sill +8.2, head +9.4 a washable insect / dust screen inside; a closable damper (motorised with a manual override, closed in dust storms and cold weather); a cap roof with a drip; an access hatch in the south face at roof level.
- **Shaft:** the tower's RC walls, clear ≈ 0.78 × 1.3 m; steel rungs on the east wall from the head hatch to the base; the ground-floor slab is a **grating** inside the shaft (the second access hatch, A5).
- **Base chamber:** the shaft walls continue to −3.15 as one RC box (the foundation of the tower, of C24 / C30 and the ends of W3 / W4: engineer, A1). Floor −2.8.
- **Tunnel:** a separate waterproof RC box, outer 1.4 m wide, top of the roof slab −1.3, underside −3.15 clear 1 × 1.4 m (a person bent over); floor falling 1.5 % from −2.8 at the tower to −2.9 at the chamber; a flexible joint with a waterstop at each end (structurally independent of the house); not insulated from the soil (ground cooling wanted), waterproofed outside, epoxy inside with coved corners.
- **Water channel:** the last 3.5 m before the chamber, 0.40 wide, 0.08 deep water (0.12 channel), stainless steel or tiled, recirculated from a 60 L reservoir in the chamber; a timer drains and dries it every day. **No sprays, no mist: rule.**
- **Outlet chamber:** RC box in the courtyard floor against the north wall, clear 1.15 × 1.60, floor −2.9, sump 0.4 × 0.4 × 0.4 with a small pump; grating hatch 0.6 × 0.9 and a fixed ladder; the outlet a louvred kerb 0.45 m high on the courtyard side with a damper; a place for a removable desert cooler (socket, water point, drain to the sump).

Long section along the air path · tower head → shaft → base chamber → tunnel → water channel → outlet chamber → courtyard

unfolded along the axis (the bend at x 7.27 is straightened) · levels from the finished ground floor ±0.00 · all assumed



How it works (qualitative only)

- **Catch:** the head stands 2.5 m above the roof in the undisturbed north-west wind. Openings on the north and west faces; the south and the cut (north-east) faces are closed, so the head does not take air from the leeward side. Louvres keep rain out, the screen keeps insects and coarse dust out, the damper closes the system.
- **Cool:** the air falls down the shaft (the shaft and the tunnel walls are in contact with the ground, which stays cooler than the afternoon air), runs through the tunnel, then passes slowly over the water channel: some evaporation, and the wet surfaces trap fine dust. **Not calculated:** air speed, temperature drop and the share of wind-driven versus stack-driven flow are for the MEP engineer.
- **Deliver:** the air enters the courtyard low, through the louvred kerb at the north wall; the courtyard is enclosed and shaded, so the cool air pools there and reaches the rooms around it through their openings; warm air leaves through the open sky.
- **Operate:** open in the evening, the night and the early morning; closed in hot afternoons, dust storms and cold weather. The channel runs only while the system is open and is **drained and dried every day.**

Health rules (stated, not optional)

- **No sprays, no mist, no fogging:** warm, stagnant or sprayed water grows Legionella and other bacteria. The channel is an open, shallow, moving surface; the air passes above it, it is never atomised.
- **No standing water:** the channel drains daily, the tunnel floor falls to the sump, the sump is pumped empty; make-up water through a float valve with a backflow preventer.
- **Smooth, washable surfaces:** epoxy or stainless steel, coved corners, no ledges; the tunnel is dark, which limits algae; a hose point in the chamber.
- **Separate from the house:** a waterproof box with joints; no house drain crosses it; the courtyard outlet closes.

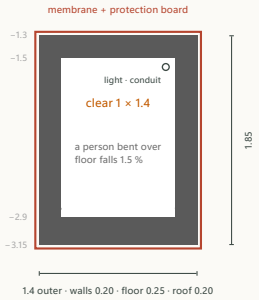
Levels along the path

head top	+9.8
head openings	+8.2 … +9.4
roof finish / roof slab	+7.30 / +7.20
ground floor finished / grating in the shaft	±0.00 / −0.08
tunnel roof top / underside	−1.3 / −1.5
tunnel floor at the tower / at the chamber	−2.8 / −2.9
box underside	−3.15
sump	−3.3
assumed pad footing underside	−1.50 (the tunnel top is above it: only at the tower exit and the chamber do footings and box meet, A1 / A3)
assumed tie beams	−0.28 … −0.90 (0.40 m above the tunnel roof)

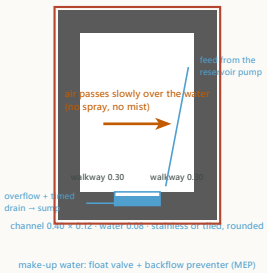
Cross sections · the tunnel, the water channel, the outlet chamber and its access

all sizes assumed · waterproofing, joints and the box design by the structural engineer; pumps, valves, timer, cooler points by the MEP engineer

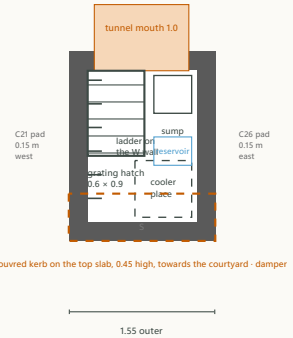
Tunnel · dry section



Tunnel · the last 3.5 m with the water channel



Outlet chamber · plan · 1.15 × 1.60 clear



Outlet chamber and access

- **Where:** in the courtyard floor against its north wall, x 6.5–8.05 (outer), between the assumed pads of C21 and C26 (0.15 m to each), east of the first-floor projection line (x 5.6) so it stands under the open sky; the colonnade column C21 (5.6, 6.5) is 0.9 m west of it; the courtyard floor drain goes to the south end, away from it.
- **Access:** a grating hatch 0.6 × 0.9 in the top slab (lockable), a fixed ladder on the west wall down 2.8 m to the chamber floor (3–4 steps cannot reach −2.9 a ladder does); from the chamber a person enters the tunnel mouth (1.0 × 1.4) and walks bent over to the tower base (the water channel has 0.30 m dry walkways on both sides). The second access is the grating in the shaft floor, reached by the rungs from the roof hatch.
- **Outlet:** a louvred kerb 0.45 m high on the courtyard side of the top slab, with an insect screen and a closable damper (the "low closable opening" of the concept). It stands in front of the east end of the women's majlis glass (x 5.1–8.1): acceptable, or the glass is shortened to x 5.1–6.4 (A8).
- **Services in the chamber:** the sump 0.4 × 0.4 × 0.4 with a submersible pump (discharge to the east manhole through a 50 mm pumped line laid back along the tunnel and out at the tower base, A7); the 60 L reservoir with the circulation pump and the timer valve for the channel; make-up water (float valve, backflow preventer); a hose point; a weatherproof socket and a light; the cooler place (0.6 × 0.6) with socket, water and a drain to the sump.

Tunnel box (structure, assumed)

- RC box, walls 0.20, floor 0.25, roof 0.20; outer 1.4 wide, top −1.3, underside −3.15 **structurally independent**: a flexible joint with a waterstop where it meets the tower base box and the chamber; movement joints with waterstops if the engineer wants more; waterproof membrane and protection board outside; the bedding and the backfill as the soil report says (X9).
- It passes 0.40 m under the assumed ground tie beams and keeps clear of the assumed pads, except at the two ends (A1, A3).

Maintenance, performance note, coordination and open items

Maintenance table (what · how often · how to reach it)

Item	What to do	How often	How to reach it
head louvres + insect / dust screen	brush and wash the screen; clear the louvres; check the drip	monthly in the dust season, else every 3 months	from the roof, through the hatch in the head's south face
damper (head)	check the motor and the manual override, grease the hinges, test closing	every 6 months and before the dust season	from the roof hatch
shaft walls and rungs	brush the walls, check the rungs and the lighting	yearly	rungs from the head hatch down to the base
shaft-floor grating (hatch 2)	lift and clean, check the lock	yearly	from the shaft
base chamber and tunnel floor	sweep and hose down; the water runs with the fall to the sump	every 3 months	from the outlet chamber (ladder), walking bent over; or from the shaft
water channel	rinse and wipe; check the feed and the overflow; the timer drains and dries it every day	weekly in use; daily automatic drain	from the chamber end of the tunnel
reservoir, circulation pump, float valve, backflow preventer	empty and clean the reservoir; check the pump and the valves	monthly in use; before the season	in the chamber
sump and pump	clean the sump, test the pump and the float switch, flush the pumped line	every 3 months	in the chamber
outlet kerb: louvres, screen, damper	wash the screen, clear the louvres, test the damper	monthly in use	from the courtyard, standing
grating hatch and ladder	check the lock, the hinges and the ladder fixings	yearly	from the courtyard
removable desert cooler (optional)	install for the season, clean the pads and the tray, remove and store	start / end of the season; pads monthly	in the chamber, through the hatch
joints and waterproofing	look for damp, cracks, salts on the walls; seepage after rain	after every heavy rain; yearly	walk the tunnel

Expected performance (qualitative only; nothing calculated)

- On nights and mornings with a north-west breeze the courtyard receives air that has passed through the ground and over the water: cooler than the street air, cleaner than ground-level air, and low, where people sit. The effect depends on the wind, the ground temperature and the humidity; the MEP engineer estimates the flow and the temperature drop and may add a low-energy fan in the shaft if the wind alone is too weak.
- Without wind the tower works weakly as a stack (cool, heavy air sinks); the optional desert cooler in the chamber gives a forced, cooled flow on still evenings.
- It is not an air-conditioning system: it cools the courtyard and the rooms around it when they are open to it, and it needs the daily routine of opening, closing and draining.

Coordination

- **Structure (studies/structure.typ):** the tower's RC walls are part of the east stiff box (W3 / W4 end); the base box replaces the pad of C24 and carries the W3 / W4 ends (A1); the tunnel keeps clear of the assumed pads of C25, C26 and C21 and passes 0.40 m under the tie beams; the women's majlis south wall sits on a ground beam that bridges the tunnel mouth at the chamber (A3); the chamber is a box in the courtyard slab between the C21 and C26 pads.
- **Plumbing:** the kitchen, laundry and maid-bath drains run east to the east manhole and never cross the tunnel; the sump's pumped line runs back along the tunnel and out east at the tower base (A7); the courtyard floor drain at the south end of the courtyard (A6).
- **Courtyard design:** the chamber and its kerb stand against the north wall, east of the projection line, under the open sky; the colonnade column C21 stays 0.9 m away; the kerb in front of the glass's east end (A8).
- **Electrical:** damper motor and lighting in the head and the shaft, light and sockets in the chamber, the two pumps, the timer valve: one dedicated circuit with an RCD (MEP).

Open items for the engineers

A1	structure	The tower base box (−0.08 … −3.15) as the foundation of the tower, of C24 / C30 and of the ends of W3 / W4; the tunnel leaves it through the W3 line: design the box and the crossing.
A2	structure	The tunnel box: section, reinforcement, joints with waterstops, bedding, backfill, uplift / groundwater; soil report first (X9).
A3	structure	The women's majlis south wall (y 6.5) over the tunnel mouth: a ground beam bridging the box between the C21 and C26 pads, independent of the chamber.
A4	MEP	Airflow: expected flow and temperature drop (wind-driven and stack); whether a low-energy fan is needed; the head opening sizes; the damper control (wind, dust, temperature sensors or manual).
A5	MEP / safety	The shaft as the second access: rungs, fall protection, lighting; a confined-space rule for the tunnel (two people, ventilation open while cleaning).
A6	MEP	The courtyard floor drain at the south end, away from the chamber; rainwater falling into the kerb outlet: a drained tray inside the kerb.
A7	MEP	The sump's pumped line to the east manhole (50 mm, along the tunnel, out at the tower base, under the maid bath, shallow); or a gravity drain if the manhole invert allows.
A8	owner / design	The outlet kerb in front of the east end of the women's majlis glass (x 5.1–8.1): accept, or shorten the glass to x 5.1–6.4.
A9	MEP	Water quality: make-up through a float valve and a backflow preventer; the daily drain-and-dry cycle; a monthly reservoir clean; no sprays or mist (rule).
A10	structure / 3D	The roof and first-floor openings for the shaft (framed by B15 / R9), the head above the roof (2.5 m, wind load), the cap and the hatch.

Sizes assumed: head 2.5 m above the roof, openings 0.8 wide × 1.2 high on the north and west faces; shaft ≈ 0.78 × 1.3 clear; tunnel clear 1.0 × 1.4 in a 1.4 m box, top −1.30, fall 1.5 %; channel 0.40 × 0.12, water 0.08, 3.5 m; chamber 1.15 × 1.60 clear, floor −2.90, sump −3.30; kerb 0.45; pads 1.5 × 1.5 with underside −1.50, tie beams to −0.90. Nothing here replaces the MEP or structural design.

✓ Air system study · route A under the women's majlis · tunnel axis 7.53 m · water channel 3.5 m · 10 open items for the engineers